linkedin.com/in/aragya-goyal/
github.com/Aragya1642

Aragya Goyal

3209 Joe Hammer Sq, Pittsburgh, PA

Mobile: +1-610-615-7007
Email: agoyal1642@gmail.com

Education

- **University of Pittsburgh (Swanson School of Engineering/Frederick Honors Col.)** **Pittsburgh, PA**
  B.S. - Computer Engineering (Autonomous Systems Focus); **GPA: 3.99/4.00** *August 2022 - April 2026*
- **Spring-Ford Area High School** **Pittsburgh, PA**
  High School Degree (Summa Cum Laude); **GPA: 100.88/100.00** *August 2018 - June 2022*

Professional Experience

- **ESTAT Actuation** **Pittsburgh, PA**
  Robotics Developer Co-op *May 2025 - Present*
  - Building and setting up four new robotic test stands to improving testing capability by 133%.
  - Designing PCB for 3 different customer facing demos, making it easier to debug issues as they arrive.
  - Testing new generation rotary clutch prototypes, creating documentation explaining all the steps taken, and using these prototypes to inform design decisions for next generation production clutches.
  - Developing a PCB for a lightweight electroadhesive exotendon system with potential applications in military and surgical uses.
- **Carnegie Mellon University Robotics Institute** **Pittsburgh, PA**
  Undergraduate Roboticist Intern *April 2023 - Present*
  - **ZOË 2 Rover**: (tinyurl.com/zoe2rover)
    * Developing low-level software stack for the 2nd generation Zoë Rover set to conduct science in the Atacama Desert including CANopen protocols via the ros_canopen package for ROS2 to communicate with encoders and motors.
    * Conducted motor and motor controller datasheet specifications validation via physical testing.
  - **Underwater Snake Robot**: (tinyurl.com/humrsCMU)
    * Implemented High-Frequency Injection methods in Brushless DC thrusters to achieve control at low/zero speeds thus reducing minimum speed by 80% allowing for reduced oscillations in station-keeping control loops.
    * Deployed station-keeping feature using AprilTags, IMU readings, and Nested PID Controllers to perform robot state-estimation underwater.
    * Conducted major repairs on the buoyancy module of the robot and assisted in continual electrical maintenance.
  - **Apple's E-Waste Recycling Project**: (tinyurl.com/applecmu)
    * Created datasets consisting of thousands of images for Machine Learning Models to detect screws in e-waste images.
    * Integrated ROS1 and Python packages to track AprilTags using a Realsense camera for localization of UR3 robotic arm.
    * Manufactured custom AprilTags using lasercutters and sheet metal manufacturing methods.

Student Organizations

- **Robotics and Automation Society (University Rover Challenge)** **Pittsburgh, PA**
  Chief Engineer/Integration Lead *August 2022 - Present*
  - Leading a multidisciplinary team of 30 students in engineering a rover, coordinating integration efforts across mechanical, electrical, software, and science teams. (tinyurl.com/roverimages)
  - Proposed and spearheaded the development of a prototype robotic hand utilizing pneumatics. (tinyurl.com/hydraarm)
  - Securing and managing over $7,000 in funding to support team development and future growth.
- **Formula SAE - Panther Racing** **Pittsburgh, PA**
  R&D Engineer *August 2022 - March 2024*
  - hello
- **FIRST Robotics** **Exton, PA**
  Team Captain/Lecd Mechanical Design Engineer *January 2020 - August 2022*
  - Led a team of 40 students and qualified for the Worlds level of competition, the highest win percentage since 2005, and a top 5% ranking globally. (FIRST Link: https://tinyurl.com/dewbot17)
  - Utilized Solidworks to design and develop competition and award-winning robots.
- **VEX Robotics** **Royersford, PA**
  Team Captain *August 2018 - June 2022*
  - Organized VEX robotics competition event for 60+ teams, hosted workshops to teach CAD to fellow club members, and received recognition for outstanding leadership. (VEX Link: https://bit.ly/3w7a6b1)
  - Qualified for the State's level of competition for all 4 years of high school.

Projects

- **555 Timer Roulette Board (KiCAD, Onshape, Circuit Design)**: Designed and built a PCB-based LED Roulette Wheel using a 555 timer and CD4017 counter, featuring dynamic speed control through capacitive timing, dip switch power control, and a custom 3D-printed enclosure for portability.(tinyurl.com/555roulette)
- **Bop It Project (KiCAD, Onshape, Microcontrollers)**: Developed an interactive "Space Invaders Bop-It" reflex game using a bare ATmega328P chip, integrating analog inputs, audio-visual feedback, and custom embedded logic for randomized commands and progressively challenging gameplay. (tinyurl.com/spacebopit)
- **Micromouse Project (Circuit Design, Robotic Control)**: Designed, assembled, and programmed a custom Micromouse robot through iterative PCB development, integrating motor control, IMU, IR sensors, and a display system with low-level embedded programming. (tinyurl.com/goyalmm)
- **Trains Project (Systems and Software Engineering)**: (tinyurl.com/agtrain2025)
- **VoltVitals Embedded Board**: (tinyurl.com/voltvitals)
- **An Artistic take on Bipedal Robots (Onshape, Arduino)**: Modified a preexisting bipedal robot to carry a symbolic lantern given to all freshmen at the University of Pittsburgh, reimagining a campus tradition through an artistic and technological lens for a humanities course on the intersection of art and technology. (tinyurl.com/gbipedal)
- **Calico AI Event Curator (AI, Website Development)**: (tinyurl.com/steelcalicoai)
- **STM-32 Elevator Simulator (ARM Assembly, Project Integration)**: Designed and Implemented software architecture in ARM Assembly to operate a physical elevator simulator PCB. (tinyurl.com/stmelevator)
- **Custom Cane (Human Centered Design, Solidworks)**: Designed and fabricated a Walker-Cane Fusion to increase bathrooms accessibility for wheelchair users. Won first place at the Senior Design Expo. (tinyurl.com/CustomCane)
- **Formula SAE E-Brake Bias (Solidworks)**: Developed an e-brake bias system for a formula style racecar, utilizing Solidworks and 3D printing technology to enhance performance and usability. Won the FSAE Innovation Award for the design and implentation of the project. (tinyurl.com/ebrakeb)
- **Formula SAE Low-Cost Slip Angle Sensor (OpenCV, Raspberry Pi)**: Worked to design and code prototypes of sensors which would allow for validation of slip angle using mouse sensors, digital cameras, and IMU's.
- **String Art Generator and Optimizer (Research, Python)**: Developed an innovative string art optimization tools and GUI using Python programming to improve upon existing string art generators. (tinyurl.com/goyalstring)
- **Silicon Prosthetic Hand (High School Independent Research, Solidworks)**: Designed and manufactured prototype prosthetic hand with silicone soft actuators and tested with human participants for an AP Research Project. (Link: https://tinyurl.com/myprosthetic)

Skills and Awards

- **Languages**: Python, C++, C, ARM Assembly, RISC-V Assembly
- **Technologies**: Linux, ROS1, ROS2, Docker, Github, Solidworks, KiCAD, MATLAB, Microsoft Suite, Microcontrollers, OpenCV
- **Manufacturing**: Milling, Soldering, Laser Cutting, General Shop Tools
- **Awards**: Dean's Honor List (2021-Present), Honor List (2021-Present), Eagle Scout

Publications

- M. Nye, A. Raji, A. Saba, E. Erlich, R. Exley, A. Goyal, A. Matros, R. Misra, M. Sivaprakasam, M. Bertogna, D. Ramanan, and S. Scherer, **"BETTY Dataset: A Multi-modal Dataset for Full-Stack Autonomy,"** in Proc. IEEE Int. Conf. Robot. Autom. (ICRA), 2025. [Online]. Available: (arxiv.org/abs/2505.07266)